

SE3070 - Case Studies in Software Engineering

Assignment 02 - Proposed Improved Design

UC-02: Submit and Verify Hazard Report

Improved version of the Group_026 design, Case Study 2:
Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka

Contents: 1. Use case scenario · 2. Use case diagram · 3. Class diagram · 4.
Sequence diagram · 5. Wireframe changes · 6. Change justification

1. Corrected Use Case Scenario

Use Case ID	UC-02
Use Case Name	Submit and Verify Hazard Report
Actors	Primary: Citizen, Community Volunteer (specialises Citizen), Duty Officer. Secondary: Location Service, Notification Service.
Description	A citizen or community volunteer submits a ground-level hazard report (photo, location, description, hazard type). A Duty Officer reviews it and either confirms or dismisses it. Only confirmed reports can be escalated into an official warning (UC-01, Alternative Flow A1).
Trigger	A citizen or volunteer observes a hazard indicator such as a rising river, a landslide crack or a blocked road.
Preconditions	• The reporter has a registered account with a district on file (required so the outcome can be sent back to them). • The hazard type list is configured in the system. • The Duty Officer is authenticated with verification privileges for the district.
Postconditions (success)	• A HazardReport exists with status CONFIRMED or DISMISSED, with the reviewing officer and review time recorded. • A dismissed report stores a reason code. • The reporter has been notified of the outcome. • A CONFIRMED report is eligible for escalation through UC-01 A1. Escalation is officer-initiated, never automatic.
Postconditions (failure)	The report remains PENDING, or is held on the device awaiting sync. No warning level is changed.

Main Flow

1. Reporter opens the app and selects "Report a Hazard".
2. App opens the camera and the reporter captures a photo of the hazard.
3. App requests the current position from the Location Service and tags the GPS coordinates.
4. Reporter enters a short description (max 200 characters) and selects a hazard type (Rising river/Flood, Landslide/Crack, Blocked road, Other).
5. Reporter submits the report.
6. System validates the report: photo present, hazard type selected, location within Sri Lanka, description length within limit.
7. System checks for PENDING reports of the same hazard type within 500 m and 2 hours, and finds none.
8. System stores the HazardReport with status PENDING and location source GPS, and returns a reference number. The app shows "Submitted – pending verification".
9. System notifies the on-duty Duty Officer for that district.
10. Duty Officer opens the pending reports queue and selects the report.
11. System displays the photo, map location, description, hazard type and submission time.
12. Duty Officer confirms the report.
13. System sets the status to CONFIRMED, records the reviewer and time, and marks the report eligible for escalation.
14. System notifies the reporter that the report was confirmed.
15. Use case ends.

Alternative Flows

A1 - Report dismissed (step 12)

- A1.1 Officer selects Dismiss and chooses a reason: Inaccurate, Duplicate, Not a hazard, or Insufficient evidence. An optional note can be added.
- A1.2 System sets the status to DISMISSED and stores the reason. The report is not eligible for escalation.
- A1.3 System notifies the reporter politely. Resume at step 15.

A2 - GPS unavailable (step 3)

- A2.1 The Location Service returns no fix within the timeout.
- A2.2 App prompts the reporter to set the location manually, by placing a pin on the map or entering a place name.
- A2.3 System records location source MANUAL, which is shown to the officer at step 11. Resume at step 4.

A3 - No connectivity (step 5)

- A3.1 App saves the report on the device and shows "Saved offline - will be sent automatically".
- A3.2 When connectivity returns, the Sync Service submits the report. Resume at step 6.

A4 - Possible duplicate (step 7)

- A4.1 System finds a matching PENDING report.
- A4.2 System links the new report to the same cluster instead of discarding it.
- A4.3 At step 11 the officer sees the cluster and its report count. Resume at step 8.

Exception Flows

E1 - Invalid report (step 6)

- E1.1 The photo, hazard type or location is missing, the location is outside Sri Lanka, or the description is too long.
- E1.2 System rejects the submission and highlights the invalid fields.
- E1.3 Reporter corrects them. Resume at step 5.

E2 - Sync failure (A3.2)

- E2.1 The sync fails.
- E2.2 The report stays on the device and is retried automatically. The reporter sees "Waiting to send".

E3 - Already reviewed (step 12)

- E3.1 Another officer has already confirmed or dismissed the report.
- E3.2 System rejects the update and shows the current status. Use case ends.

2. Corrected Use Case Diagram (UC-02 scope)

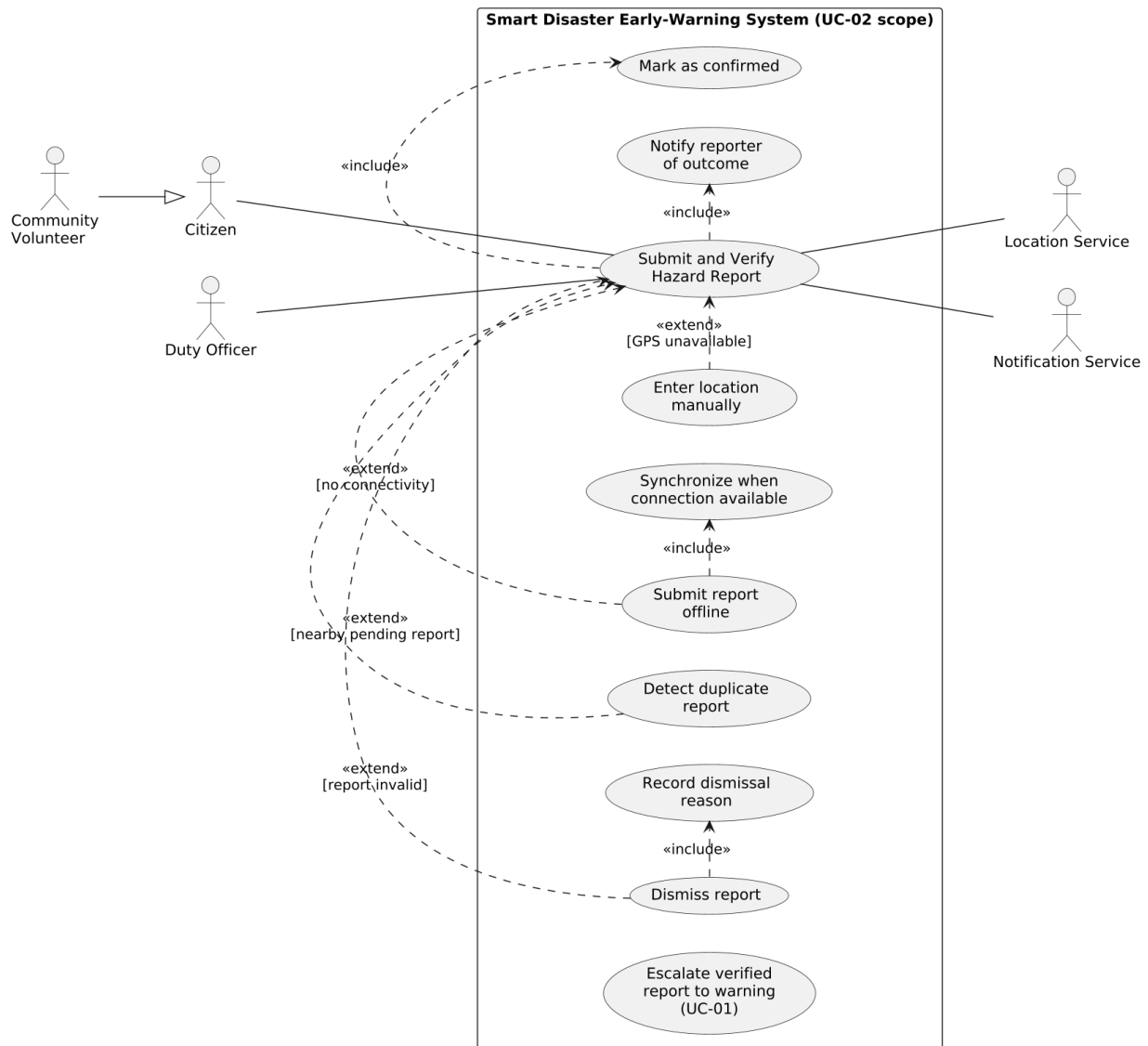


Figure 1: Improved use case diagram for Submit and Verify Hazard Report

Changes from the original

- Removed «include» on Attach photo, Capture GPS location and Select hazard type. These are steps of the flow, not use cases, so they belong in the scenario.
- Added a generalization arrow from Community Volunteer to Citizen, matching the class diagram.
- Added «extend» *Enter location manually* [GPS unavailable].
- Kept «extend» *Submit report offline*, which «include»s *Synchronize when connection available*.
- Moved «extend» *Detect duplicate report* so it extends the submission (it happens at step 7), not Review and verify.
- Kept *Mark as confirmed*, and *Dismiss report* with «include» *Record dismissal reason*.
- Added «include» *Notify reporter of outcome*.
- Connected Location Service and Notification Service as secondary actors. Escalation to a warning remains in UC-01 (officer-initiated).

3. Corrected Class Diagram (HazardReport portion)

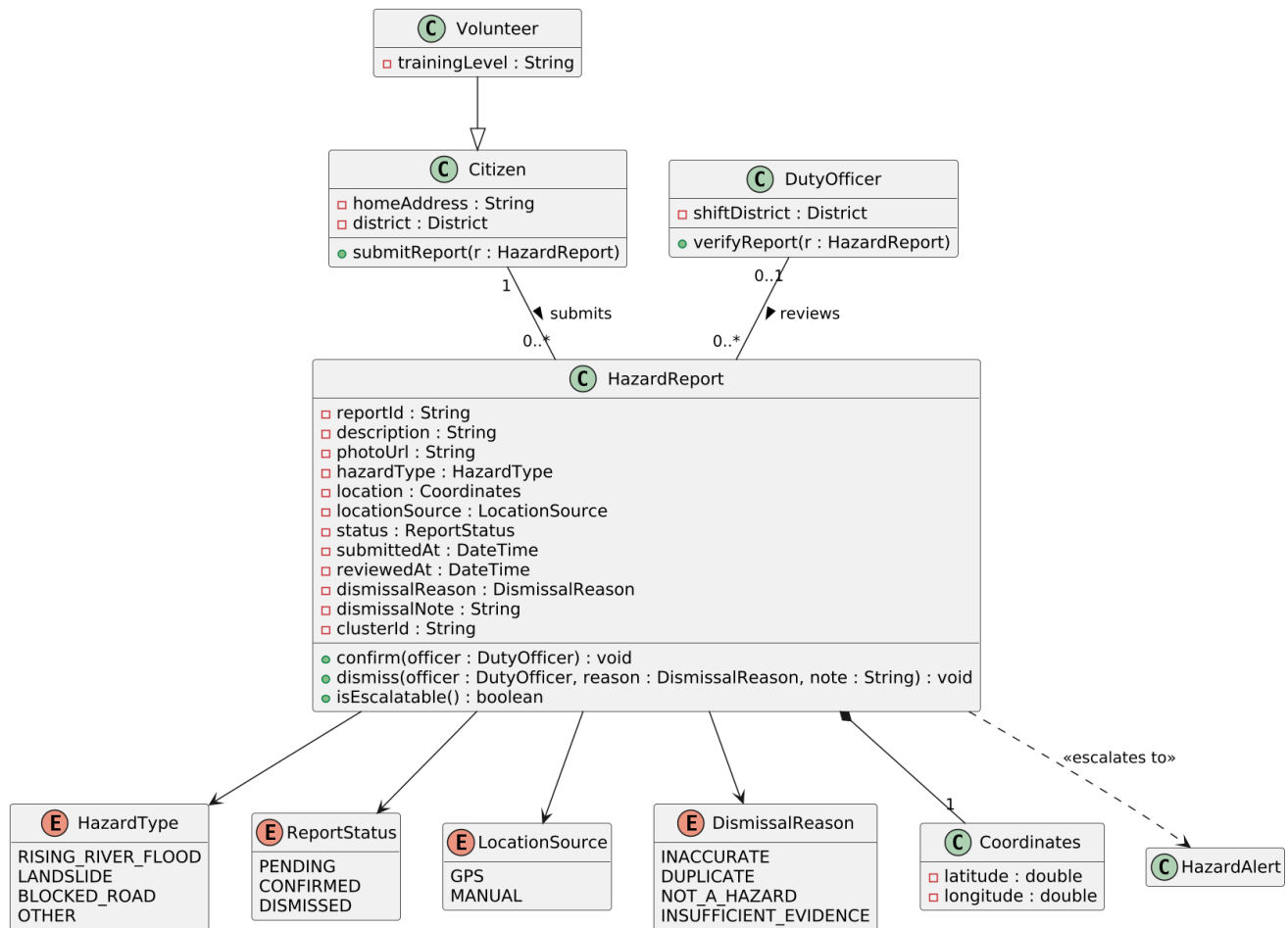


Figure 2: Improved class diagram for the hazard report domain

Changes from the original

- Added the missing **hazardType** attribute, typed by a HazardType enum.
- Replaced the String **verificationStatus** with a ReportStatus enum (PENDING, CONFIRMED, DISMISSED).
- Added **locationSource**, **reviewedAt**, the reviewing DutyOfficer association, **dismissalReason** and **dismissalNote**.
- Added **clusterId** so duplicate reports can be linked.
- Replaced the incorrect «extend» on the HazardReport–HazardAlert link with a «escalates to» dependency (extend is a use case stereotype).
- Hazard type mapping to UC-01: RISING_RIVER_FLOOD → Flood, LANDSLIDE → Landslide. BLOCKED_ROAD and OTHER are ground-impact reports the officer interprets when escalating.

4. Corrected Sequence Diagram

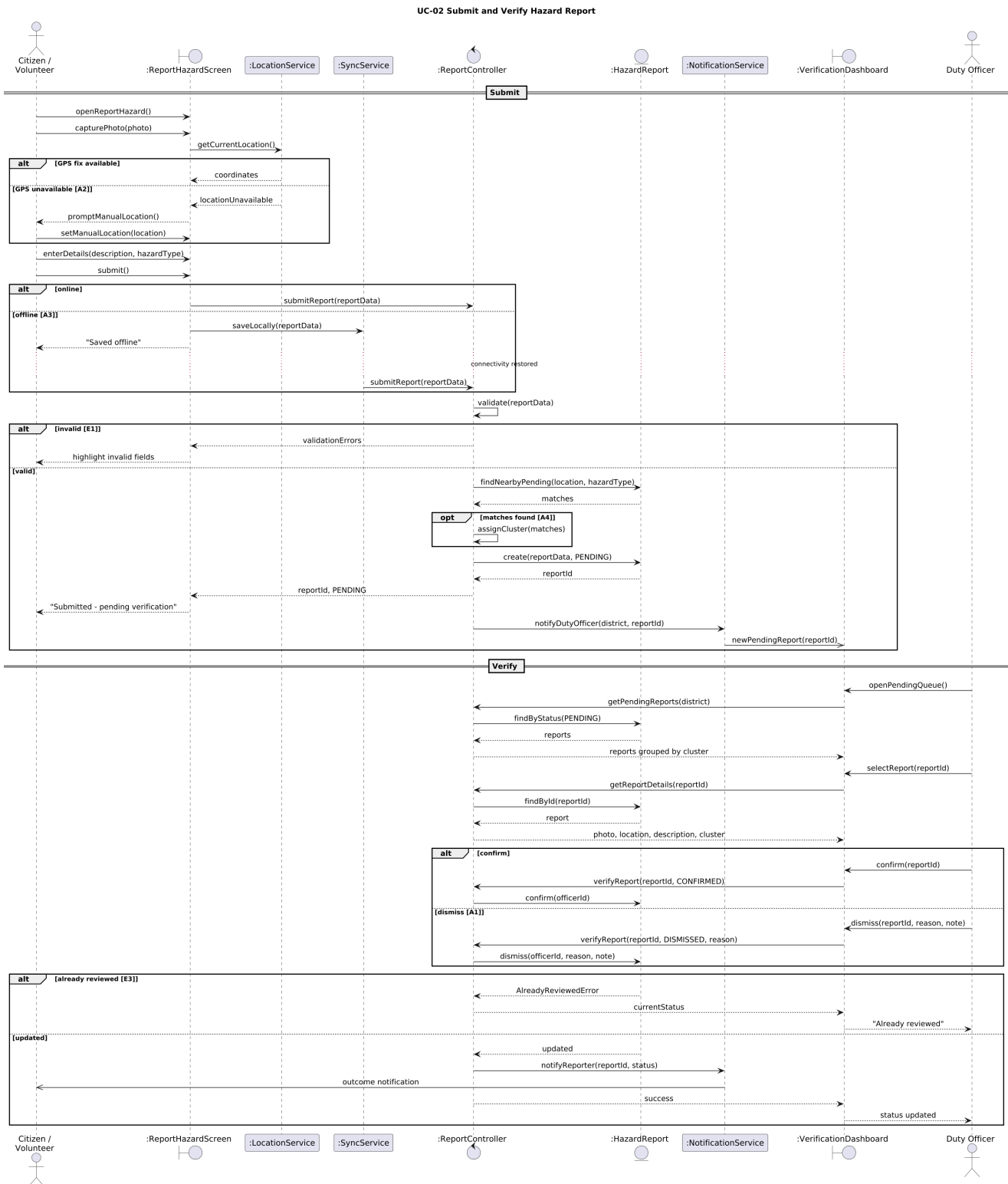


Figure 3: Improved sequence diagram – Submit and Verify Hazard Report

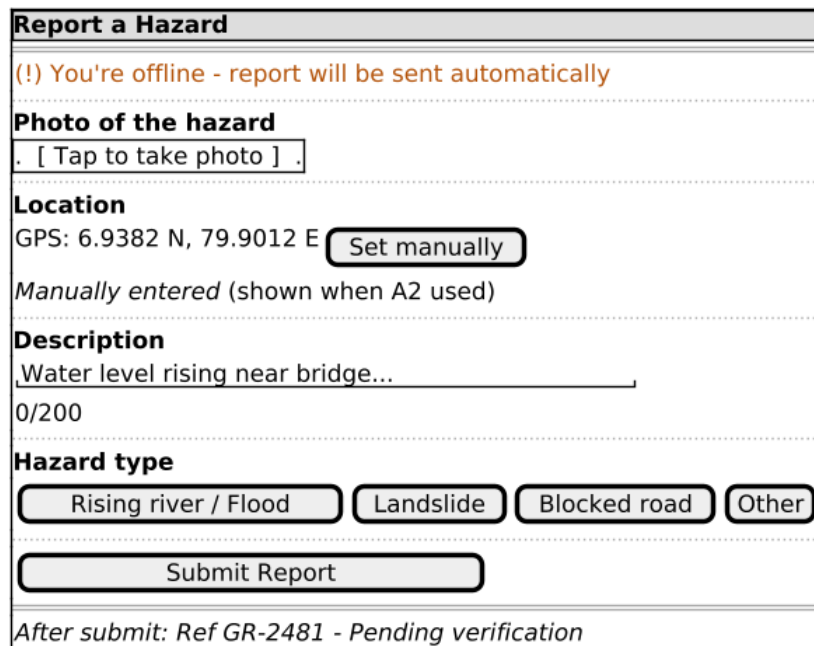
Fixes to the original

- The GPS guard is no longer inverted, and location comes from a device-side Location Service instead of a controller round-trip.
- Every message follows Boundary → Controller → Entity; the entity never calls the controller.
- Calls are solid arrows and replies are dashed (the original drew openReport() as a reply).
- A real Duty Officer actor lifeline is shown.

- Offline sync (A3), duplicate clustering (A4), dismissal reason (A1), validation (E1) and concurrent review (E3) are all shown.
- The reporter is notified in both outcomes, and the incorrect “warning level updated” reply is removed.
- Status names match the scenario (DISMISSED, not CLOSED; typo “CONFIMED” fixed).

5. Wireframe Changes

5.1 Citizen - Report a Hazard (updated)



Report a Hazard

(!) You're offline - report will be sent automatically

Photo of the hazard
[Tap to take photo]

Location
GPS: 6.9382 N, 79.9012 E [Set manually]
Manually entered (shown when A2 used)

Description
Water level rising near bridge...
0/200

Hazard type
[Rising river / Flood] [Landslide] [Blocked road] [Other]

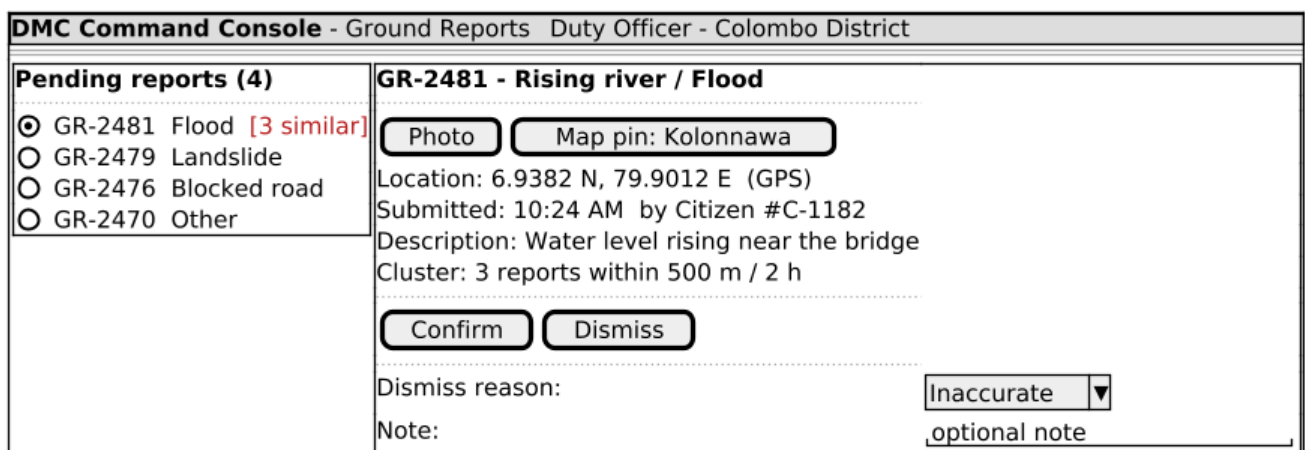
[Submit Report]

After submit: Ref GR-2481 - Pending verification

Figure 4: Low-fidelity layout of the updated citizen report screen

- Added an offline banner (A3), a “Set manually” location option and a “Manually entered” label (A2).
- Hazard chips changed to Rising river/Flood, Landslide, Blocked road and Other.
- Coordinates corrected to a valid Sri Lankan format (6.9382° N, 79.9012° E).
- Added a post-submit confirmation state with a reference number and “Pending verification”.
- Fixed the typo “Description”. The storyboard coordinates (-6.2088, 106.8456, which is Jakarta) are also replaced.

5.2 Duty Officer - Review Report (new screen)



DMC Command Console - Ground Reports Duty Officer - Colombo District

Pending reports (4)	GR-2481 - Rising river / Flood
☒ GR-2481 Flood [3 similar]	[Photo] [Map pin: Kolonnawa]
☐ GR-2479 Landslide	Location: 6.9382 N, 79.9012 E (GPS)
☐ GR-2476 Blocked road	Submitted: 10:24 AM by Citizen #C-1182
☐ GR-2470 Other	Description: Water level rising near the bridge
	Cluster: 3 reports within 500 m / 2 h
	[Confirm] [Dismiss]
	Dismiss reason: [Inaccurate ▼]
	Note: [optional note]

Figure 5: Low-fidelity layout of the new officer review screen

- Pending queue with cluster badges (e.g. “3 similar”).
- Detail panel showing photo, map pin, description, hazard type, time and location source.
- Confirm button, and a Dismiss button with a reason dropdown and optional note.
- This screen was missing from the original design; it is required so the implemented verification UI is consistent with the report.

6. Change → Justification

Change	Critique it addresses
Manual location fallback	The original reused a stale last-known location and merged GPS failure with network failure.
Offline sync in the sequence diagram	Exception E1 was in the scenario but missing from the diagram.
Duplicate check moved to submission	The use case diagram placed it at review while the scenario placed it at step 6.
Dismissal reason stored; reporter notified in both outcomes	The dismissal branch had no reason code and no notification, and used CLOSED where the scenario said DISMISSED.
BCE message flow and Duty Officer lifeline	The entity called the controller, replies were drawn as calls, and there was no actor lifeline.
Confirmation no longer changes the warning level	The original sequence reply “warning level updated” contradicted the scenario.
Enums plus hazardType in the class diagram	The hazard type was missing and statuses were loose Strings.
«extend» removed from the class diagram	A use case stereotype was misused in a class diagram.
Step includes removed from the use case diagram	Attach photo / Capture GPS / Select hazard type were functional decomposition, not use cases.
Officer review wireframe added	Half of the use case (verification) had no interface design.
Sri Lankan coordinates and hazard types aligned with UC-01	The storyboard showed a Jakarta location; report and alert hazard types were inconsistent.

Note for implementation and testing: the alternative flows (A1–A4) and exception flows (E1–E3) map directly to negative and edge-case unit tests, supporting the 80% coverage target.